TOPIC 3 · AI DETECTORS

2023–24: Detectors Didn't Work

61%

of essays by non-native English writers flagged as AI

Liang et al. 2023 · 7 detectors

14 tools

"neither accurate nor reliable"

Weber-Wulff et al. 2023 · incl. Turnitin

40% → 22%

accuracy after simple edits; 15% of human texts flagged

Perkins et al. 2024 · 7 detectors

Liang et al., Patterns 2023 · Weber-Wulff et al., Int. J. Educational Integrity 2023 · Perkins, Roe et al., IJETHE 2024

TOPIC 3 · AI DETECTORS

2025+: The Best Detectors Are Very Accurate

False positive rates of AI detectors, by text type

	Pangram	GPTZero	Originality.ai
Product review	0.5%	2.4%	1.7%
Blog post	0%	0.5%	0%
News article	0.1%	1%	0.3%
Novel excerpt	0%	0.5%	0.3%
Restaurant review	0.8%	1%	2.2%
Résumé	0%	0%	0.1%

Jabarian and Imas, 2025

≈2% or less

of AI text missed, all three tools

After a "humanizer" rewrite, missed: Pangram 0–5%, GPTZero 26–77%

Jabarian & Imas, NBER Working Paper 34223 (2025) · ~2,000 human passages vs. text from GPT-4.1, Claude Opus 4 and Sonnet 4, Gemini 2.0 Flash

TOPIC 3 · AI DETECTORS

Small Error Rates Add Up

An honest student submits 40 papers over four years. Chance of being flagged at least once:

1% false positives

33%

0.5% false positives

18%

0.1% false positives

4%

P(flagged at least once) = $1 - (1 - p)^{40}$ · assumes independent checks

DISCUSSION

Do you want to be the AI police?

A detector says a student's paper is 98% likely AI. The student says they wrote it.

What happens next, step by step?

Is that a process you want to run, all semester?

Or would you rather redesign the assignment?